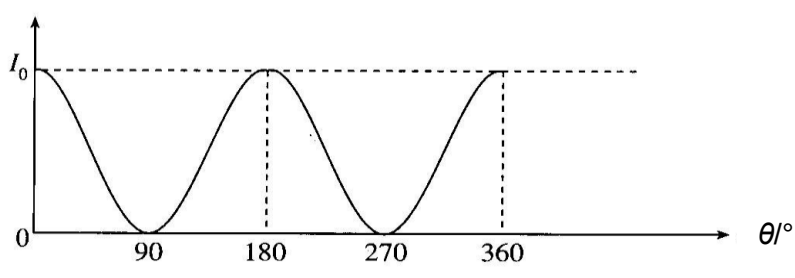
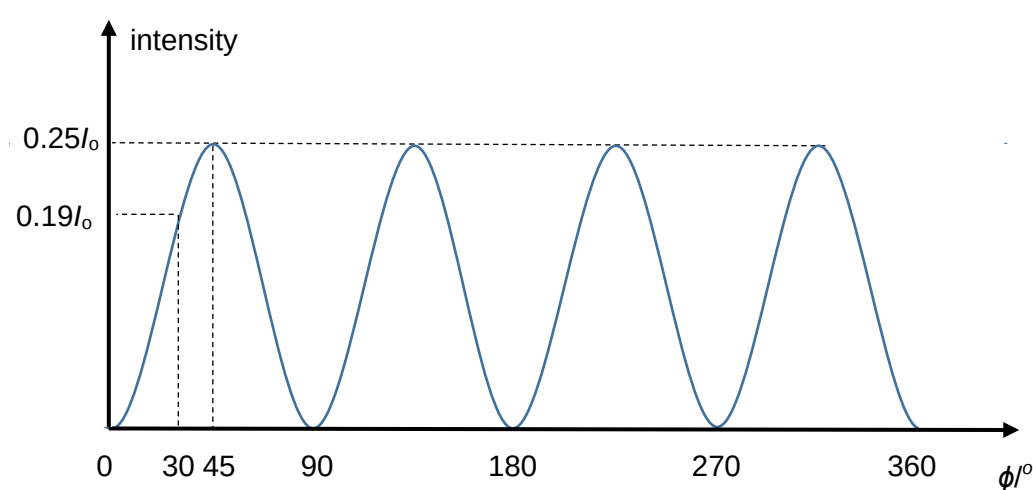
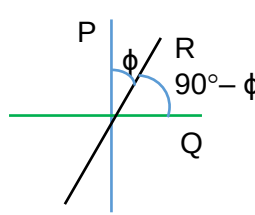


3a	Intensity is proportional to (amplitude)² Since amplitude is $A \cos \theta$, the graph of amplitude is a cosine function. Hence $I = kA^2 \cos^2 \theta$ (Malus' Law)  Shape: M1 Axes values: A1	M1 A1
3bi	$I = kA^2 \cos^2 \phi$, and $I_0 = kA^2$ Intensity after passing through polaroid R = $I_0 \cos^2 30^\circ = 0.75 I_0$ Intensity after passing through polaroid Q = $(0.75 I_0) \cos^2 (90^\circ - 30^\circ) = 0.19 I_0$	C1 A1
3bii	 <u>Working below is not required:</u> Intensity after passing through polaroid R, $I_R = I_0 \cos^2 \phi$ Intensity after passing through polaroid Q = $I_R \cos^2 (90^\circ - \phi)$ $= I_0 \cos^2 \phi \sin^2 \phi$ $= I_0 (0.5 \sin 2\phi)^2$ where $\sin 2\theta = 2 \sin \theta \cos \theta$ $= 0.25 I_0 \sin^2 2\phi$  Shape: sine-square graph with 4 cycles, starting with zero labelling of every 45° correct maximum value – $0.25 I_0$	M1 A1 B1
3c	Longitudinal waves cannot be polarised because the <u>oscillation of particles</u> in the wave is parallel to its <u>direction of energy transfer</u>.	A1

4a	progressive waves transfer/propagate energy and energy in a stationary waves is	R2
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